CÁC CÔNG NGHỆ LẬP TRÌNH HIỆN ĐẠI

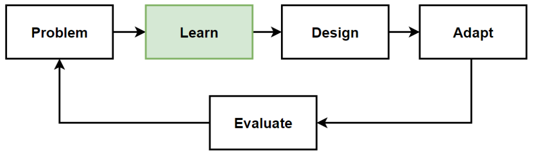
KIẾN TRÚC SERVERLES MICROSERVICE

TS. Đỗ Như Tài
Đại Học Sài Gòn
dntai@sgu.edu.vn

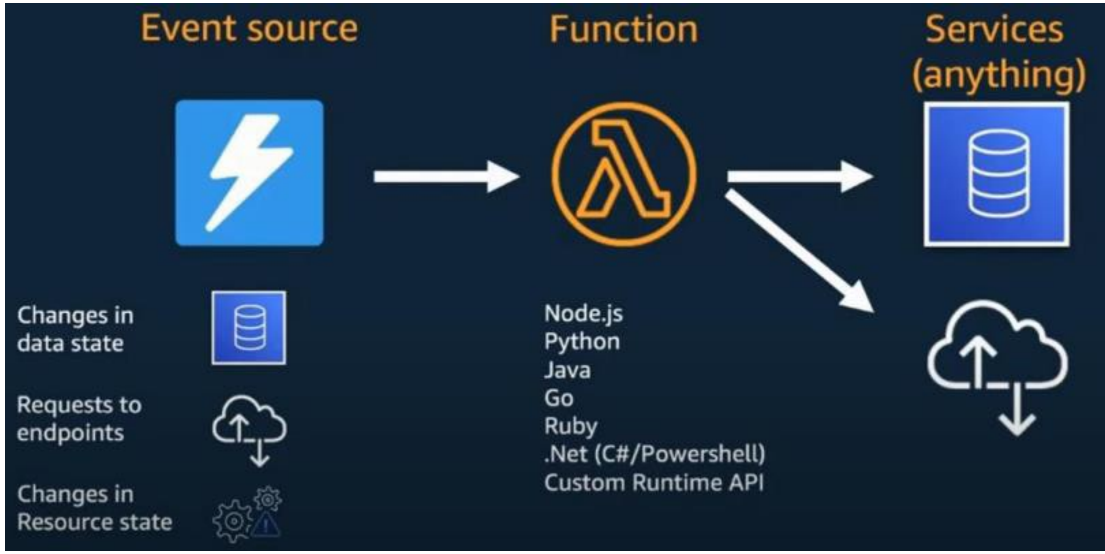
Serverless Microservices Architecture

AWS Event-driven Serverless Microservices using AWS Lambda, API Gateway, EventBridge, SQS, DynamoDB and CDK for IaC

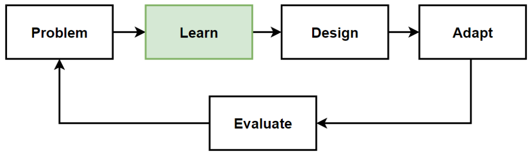
Introduction - Serverless Microservices



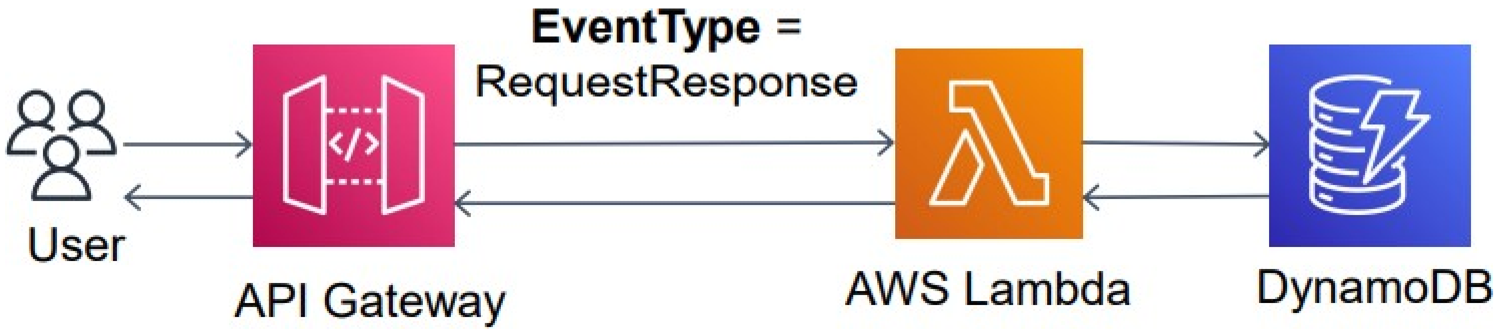
- **Serverless microservices** in which the **infrastructure** is managed by a **cloud provider** and the microservices are **executed** in a **serverless computing** environment.
- **Microservices** are **automatically scaled** to meet the **demand of the application** and developer **does not worry about provisioning** and **managing servers**.
- **AWS Lambda**, **Azure Functions** and **Google Cloud Functions** are a **serverless computing** platforms provided by AWS, Microsoft Azure and Google Cloud.
- It allows developers to **write** and **deploy microservices** that **run in response to specific events**, such as an **HTTP request** or a **change to a database**.



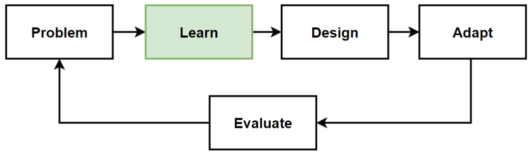
Example of Serverless Microservices



- **Write the microservice code** in a programming language supported by AWS Lambda, such as Node.js, Python, or Java.
- **Package the code** and any dependencies into a deployable package, such as a ZIP file.
- **Create an AWS Lambda function** and upload the package.
- **Configure the function** to be triggered by a specific event, such as an HTTP request.
- **Test** and **debug the function** using the AWS Lambda console or command-line tools.
- **Deploy the function** to a production environment and monitor its performance using AWS CloudWatch.



AWS Serverless services used



Compute

AWS Lambda

Databases

Amazon DynamoDB

Messaging/Application Integrations

Amazon EventBridge

Amazon Simple Queue Service (Amazon SQS)

API Management

Amazon API Gateway

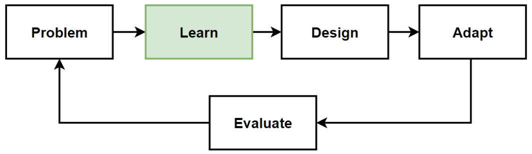
IaC

AWS CloudFormation

AWS CDK Stack

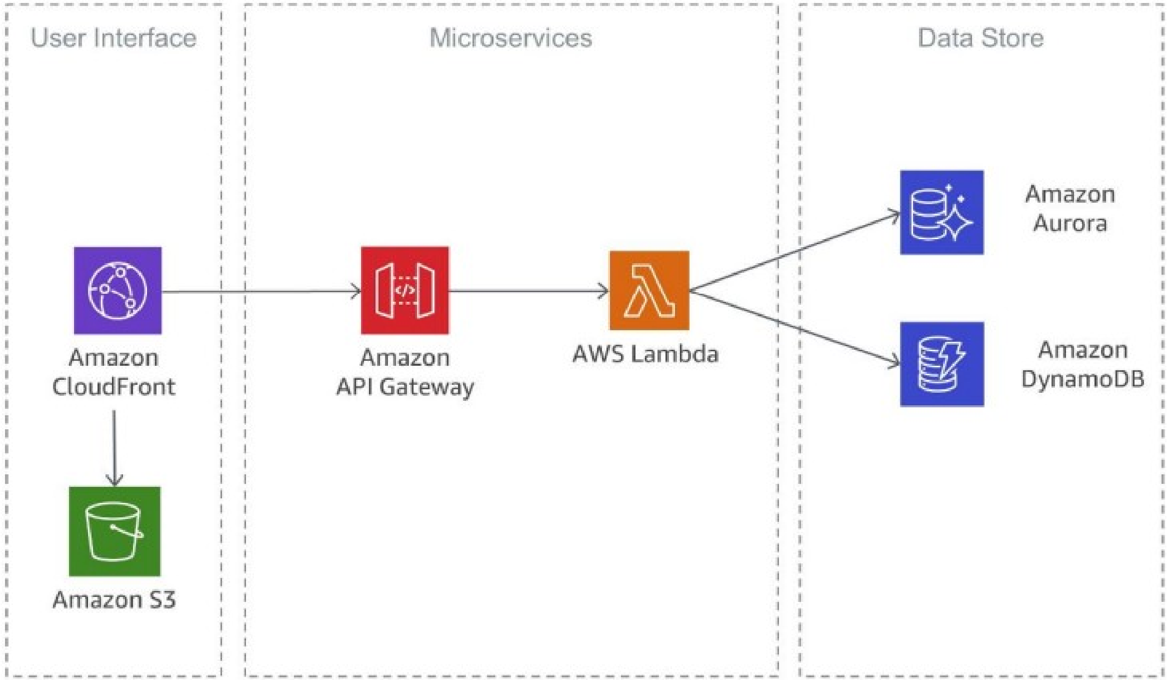
Monitoring

Amazon CloudWatch



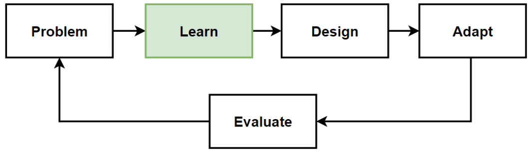
AWS Lambda as a Microservice

- Microservice are **small business services** that can work together and can be deployed **autonomously** / **independently**.
- **Lambda** is a service that allows you to run your functions in the cloud **completely serverless** and eliminates the operational **complexity**.
- It **integrates** with the **API gateway**, allows you to invoke functions with the API calls, and makes your architecture completely serverless.
- Microservice with **AWS Lambda**, which removing the architectural overhead of designing for **scaling** and high **availability**,
- Eliminating the **operational efforts** of operating and monitoring the microservice's underlying infrastructure.



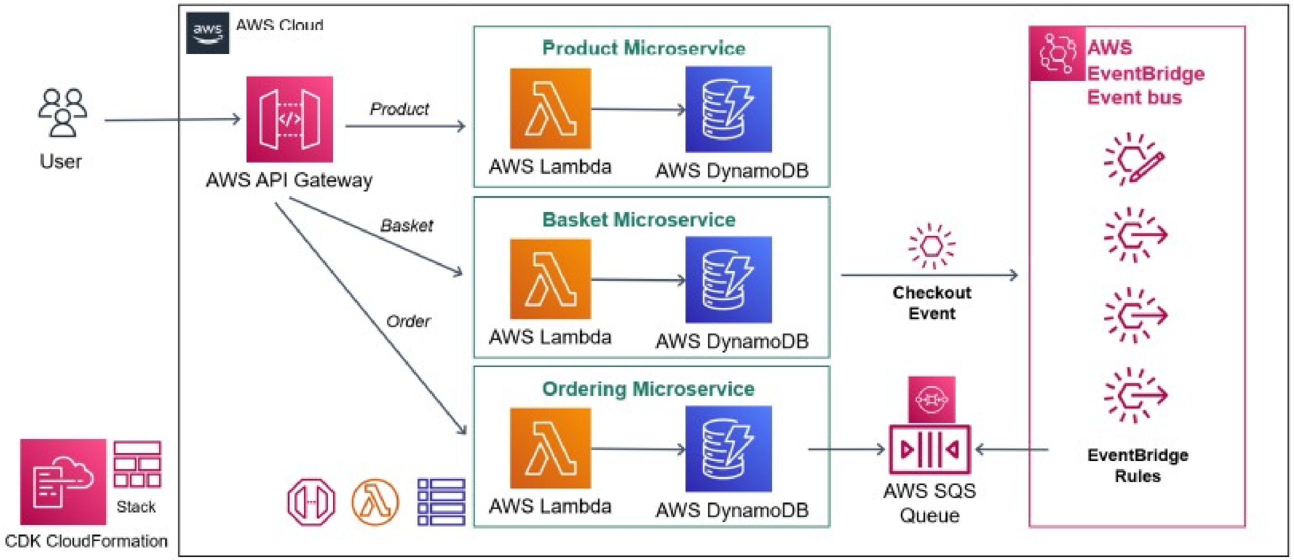
https://docs.aws.amazon.com/whitepapers/latest/microservices-on-aws/serverless-microservices.html

Serverless E-Commerce Microservice Architecture

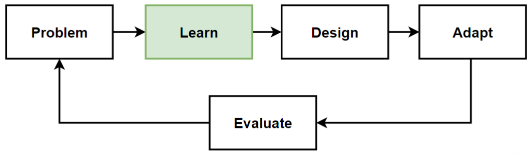


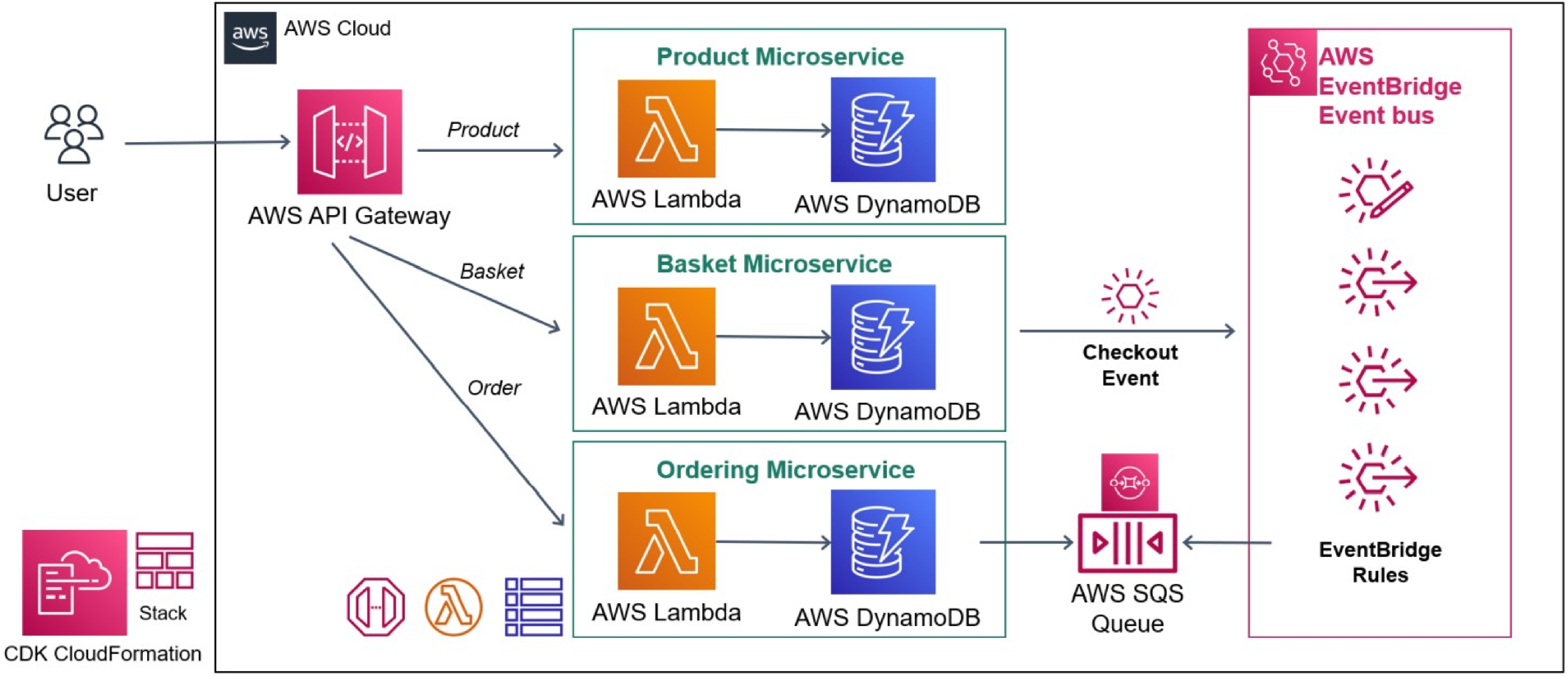
- Developed in Serverless environment
  - developing with AWS Lambda
  - No Deployment Requirement
  - and IaC by AWS CDK CloudFormation
- **AWS API Gateway**
- **Microservices** developed with AWS Lambda
- **Databases** are no-sql DynamoDB that can be

  key-value pair or document databases
- **Message broker system** which is Amazon EventBridge Eventbus
- **Amazon CloudWatch** for centralized logging and monitoring
- By default automatic scalability and high avaliability



Design: Serverless E-Commerce Microservices





CÁM ƠN ĐÃ CHÚ Ý LẮNG NGHE!